

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Supplementary Winter 2022

Course	: B. Pharmacy	Date	: 11/02/2023
Subject Name	: Physical Pharmaceutics II	Sem	: IV
Max Marks	: 75	Subject Code	: BP403T
		Duration	: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define angle of repose and give its importance.
- ii) Elaborate the concept of thixotropy.
- iii) Explain any 2 general characteristics of colloidal particles.
- iv) Distinguish between flocculated and deflocculated suspension.
- v) What is first order of reaction?
- vi) Draw HLB scale.
- vii) What measures you will take to improve the flow property of a granules?
- viii) Define gold no and Schulze hardy rule.
- ix) What is the principle of Coulter Counter?
- x) Explain the mechanism of micelles formation.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What is drug stability? Explain in details accelerated stability testing.
- ii) Define and classify colloidal dispersion. Add a note on comparative properties of different colloidal system.
- iii) Elaborate any 4 derived properties of powder.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain different factors influencing the chemical degradation of pharmaceutical product.
- ii) Differentiate between plastic and elastic deformation. Add a note on Heckel equation.
- iii) Write a note on Non-Newtonian systems
- iv) Enlist different methods to determine particle size and explain any one.
- v) Discuss sedimentation parameter.
- vi) Describe the electrical property of colloidal system.
- vii) Explain application of micrometrics in pharmacy.
- viii) Classify viscometer and explain working principle of Ostwald Viscometer.
- ix) What is emulsion? Explain stability of emulsion.

-----END OF THE PAPER-----

